CRUD Operations

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Objectives

- CRUD operations in java with JDBC.

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CRUD

- CRUD is the acronym for the following four operations.

  - C- INSERTION/CREATE
  - R- RETRIEVAL
  - U- UPDATION
  - D- DELETION

Insert Operation

```
import java.sql.*;
class AccountStoringApplication
{
public static void main(String[] args) throws ClassNotFoundException, SQLException
{
Class.forName("oracle.jdbc.driver.OracleDriver");
Connection con = DriverManager.getConnection("jdbc:oracle:thin:@localhost:1521:xe","System", “amma");
Statement st = con.createStatement();
int c = st.executeUpdate("insert into account values(1005, ‘Sita', 2345)");
System.out.println(c + "account stored successfully");
int c = st.executeUpdate("insert into account values(1006, 'kumar', 5345)");
System.out.println(c + " " more account stored successfully");
st.close();
con.close();}}
```

Retrieve Operation

```
import java.sql.*;
import java.util.*;
class AccountDetails
{
public static void main (String[]args) throws ClassNotFoundException, SQLException
{
Scanner sc = new Scanner (System.in);
System.out.println ("ENTER ACCOUNT NUMBER");
int ano = sc.nextInt ();
Class.forName ("oracle.jdbc.driver.OracleDriver");
Connection con = DriverManager.getConnection ("jdbc:oracle:thin:@localhost:1521:xe", "System", “Amma");
Statement st = con.createStatement ();
ResultSet rs = st.executeQuery ("select * from account where accno =" + ano);
```

Retrieve Operation

```
if (rs.next ())
{
System.out.println ("account no: " + rs.getInt (1));
System.out.println ("acc holder name:" + rs.getString (2));
System.out.println ("balance : " + rs.getFloat (3));
System.out.println ("address: " + rs.getString (4));
}
else
System.out.println ("account doesnot exist");
rs.close ();
st.close ();
con.close ();
}
}
```

Update Operation

```
import java.sql.*;
class UpdateAccountApplication
{
public static void main (String[]args) throws ClassNotFoundException, SQLException
{
Class.forName ("oracle.jdbc.driver.OracleDriver");
Connection con = DriverManager.getConnection ("jdbc:oracle:thin:@localhost:1521:xe", "System",
    “Amma");
Statement st = con.createStatement ();
int rows = st.executeUpdate ("update account set balance = balance+2000");
System.out.println (rows + " rows modified");
st.close ();
con.close ();} }
```

Delete Operation

```
import java.sql.*
import java.util.*;
class AccountCloseApplication
{ public static void main (String[]args) throws ClassNotFoundException, SQLException
{ Scanner sc = new Scanner (System.in);
System.out.println ("ENTER ACCOUNT NUMBER");
int ano = sc.nextInt ();
Class.forName ("oracle.jdbc.driver.OracleDriver");
Connection con = DriverManager.getConnection ("jdbc:oracle:thin:@localhost:1521:xe", "System", “Amma");
Statement st = con.createStatement ();
int c = st.executeUpdate ("delete from account where accno =" + ano);
if (c == 0)
System.out.println ("account doesnot exist");
else
System.out.println ("account closed successfully");
st.close ();
con.close (); }  }
```

Takeaways

- CRUD operations with JDBC

Reference

[1]. Java How to Program Dietel,Harvey Tenth Edition

[2].https://docs.oracle.com/javase/tutorial/jdbc/basics